

INPUT
↓

$|input| = N = 12 \Rightarrow \sqrt{N} = 3.46.. \Rightarrow \text{range of random keys} = [0, 3]$

IMPROVED WORD COUNT: **ROUND 1**

random key
↓

Doc 1
apple, orange, apple

(2, (apple, 2))
(1, (orange, 1))

Doc 2
melon, apple banana

(0, (melon, 1))
(3, (apple, 1))
(0, (banana, 1))

Doc 3
peach, melon, melon

(1, (peach, 1))
(0, (melon, 2))

Doc 4
banana, peach, apple

(2, (banana, 1))
(2, (peach, 1))
(3, (apple, 1))

(0, (melon, 1), (banana, 1), (melon, 2))

(melon, 3)
(banana, 1)

(1, (orange, 1), (peach, 1))

(orange, 1)
(peach, 1)

(2, (apple, 2), (banana, 1), (peach, 1))

(apple, 2)
(banana, 1)
(peach, 1)

(3, (apple, 1), (apple, 1))

(apple, 2)

Map Phase

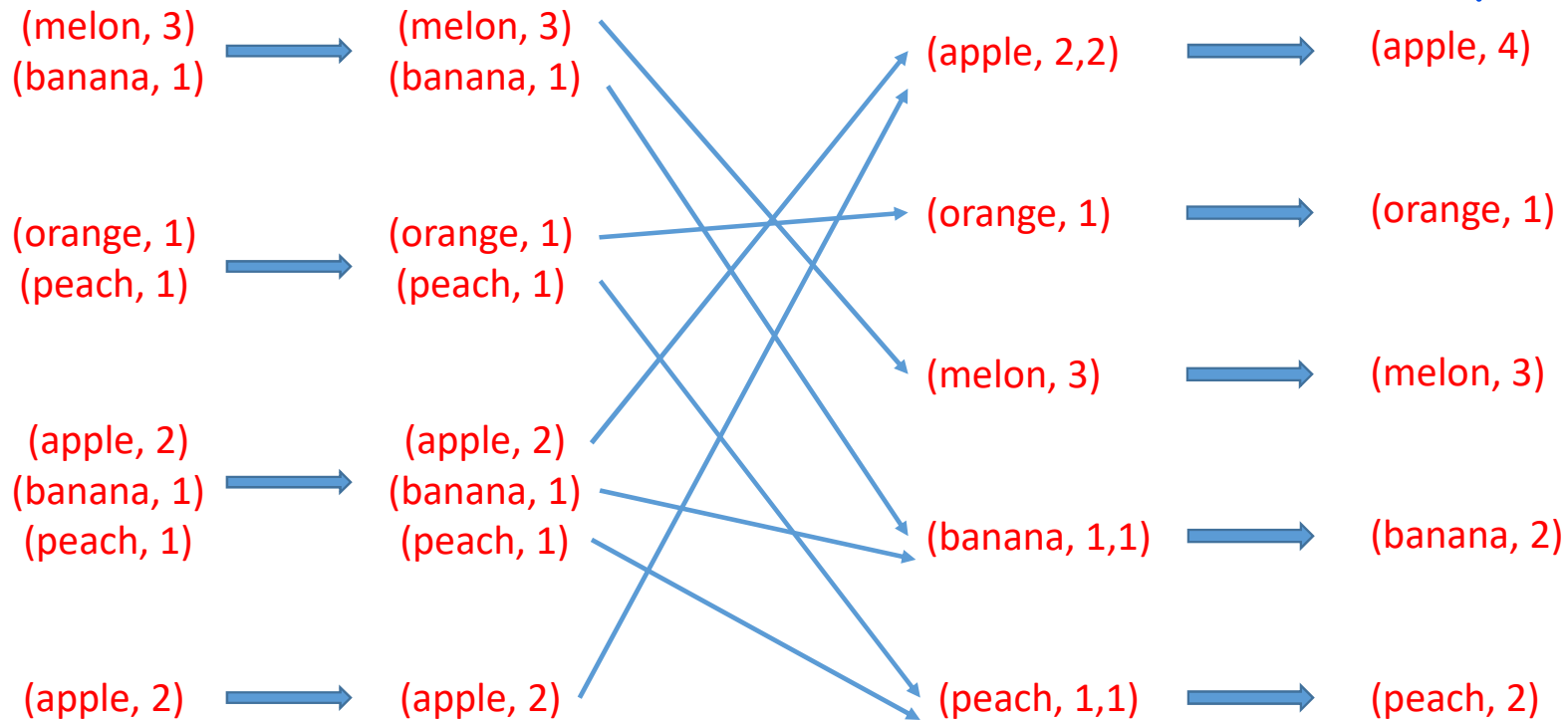
SHUFFLE

Reduce Phase

Input $\equiv$ output of Round 1

output

IMPROVED WORD COUNT: **ROUND 2**



Map Phase

SHUFFLE

Reduce Phase